

CUSTOMER BEHAVIOUR ANALYSIS

**An Analysis of Customer Segmentation,
Purchase Patterns and Churn Risk**

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Tools Used:

Python | Pandas | Matplotlib | Seaborn | Google Colab

1. Executive Summary

This project analyzes customer transactions and purchasing behavior to identify customer segments, purchase patterns, retention trends, and churn risks for Alfido Tech. The dataset contains 250,000 transactions from 49,661 unique customers. The analysis involved data cleaning, exploratory data analysis, feature engineering, RFM analysis, customer segmentation, churn analysis, and retention analysis.

The Customer Behaviour Analysis project focuses on analyzing customer purchasing patterns, segmentation, and value contribution using data-driven techniques. The objective of this project is to understand customer behaviour, identify valuable customer segments, and generate actionable business insights that can help improve customer retention, marketing strategies, and revenue growth.

RFM analysis classified customers into five segments: Champions, Loyal Customers, Potential Loyalists, At Risk, and Lost Customers. Potential Loyalists represent the largest segment with 13,357 customers, while Champions have the highest average monetary value of 22,506.30. The repeat customer rate is 96.67%, indicating strong repeat-purchase behavior. However, Champions also have the highest churn rate at 20.59%, making their retention an important business priority.

Based on these findings, segment-specific engagement, loyalty, and churn-prevention strategies are recommended for Alfido Tech.

2. Dataset & Methodology

The analysis was performed using Python in Google Colab. Pandas was used for data manipulation and analysis, while Matplotlib and Seaborn were used for data visualization.

The dataset contains 250,000 transactions and 13 original columns. Data preparation included checking data types, missing values, duplicate records, redundant columns, and numerical ranges.

Additional date and age-group features were created before performing RFM analysis. RFM analysis was used to evaluate customers based on Recency, Frequency, and Monetary value and to create meaningful customer segments.

Dataset Overview

Total Customers: 49,661

Total Transactions: 250,000

Total Revenue: 681,346,299

Average Purchase Amount: 2,725.39

Repeat Customer Rate: 96.67%

3. Customer & Purchase Analysis

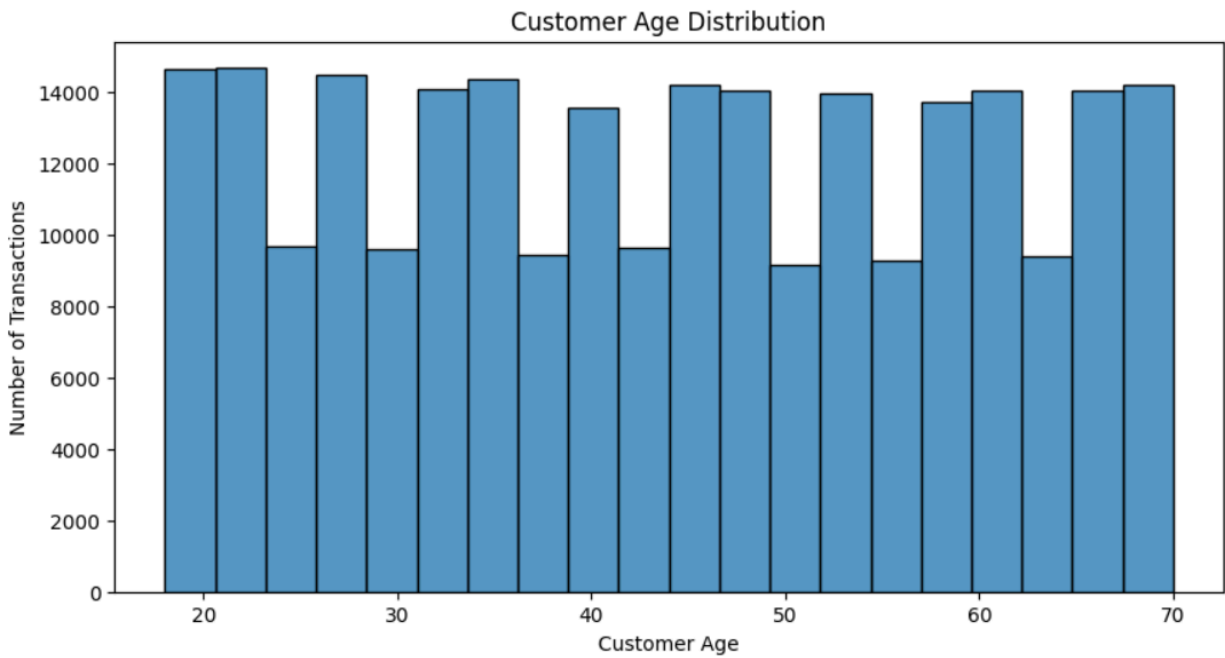


Figure 1: Customer Age Distribution

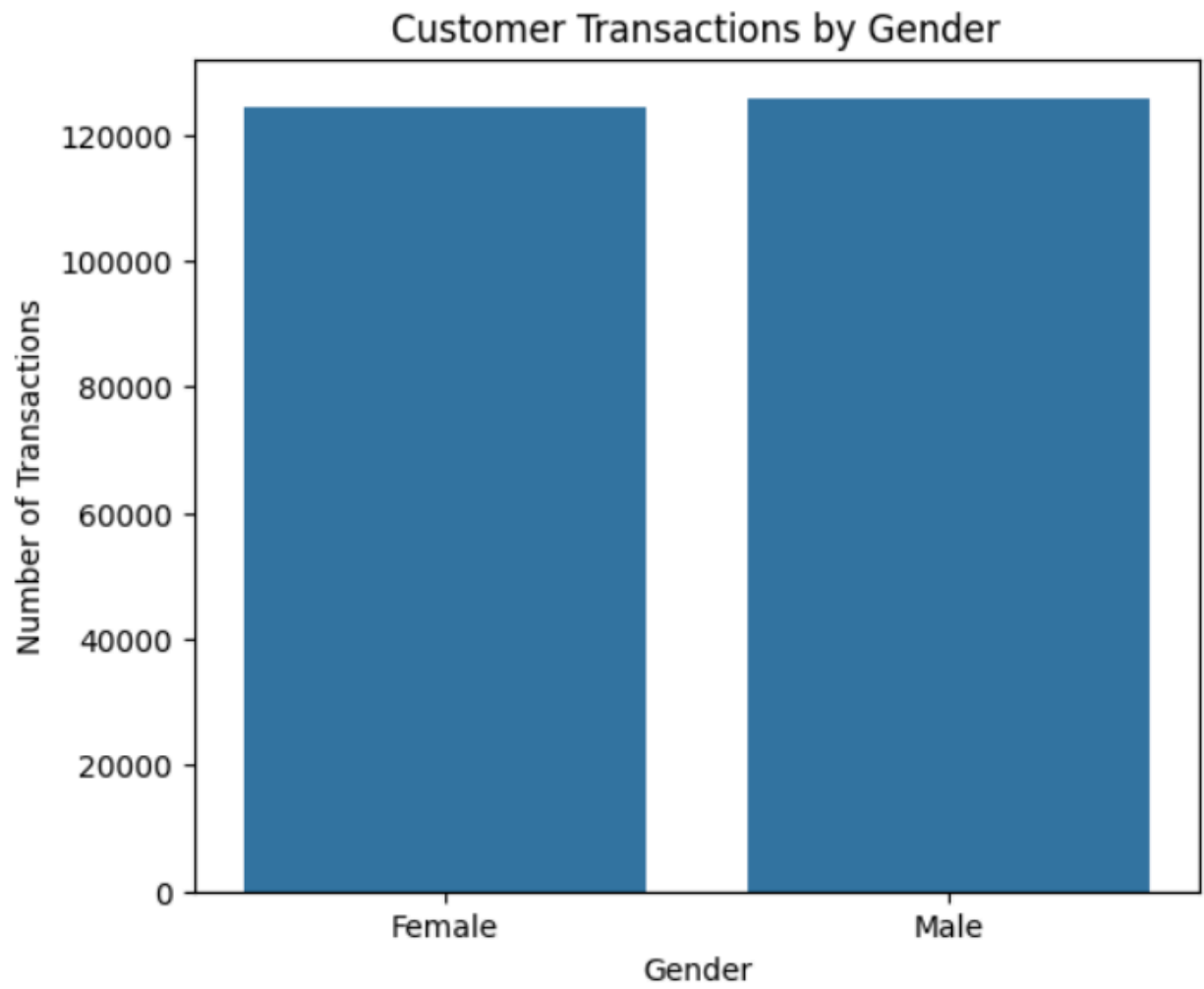


Figure 2: Transactions by Gender

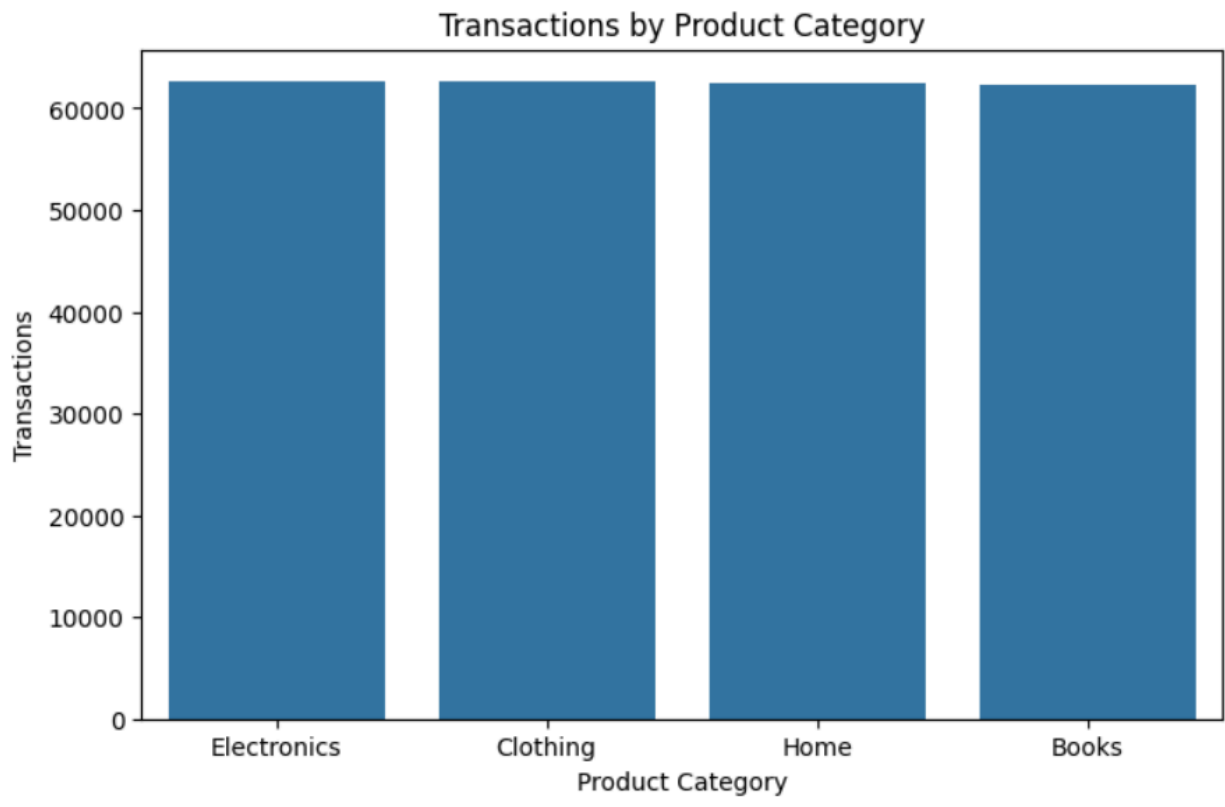


Figure 3: Transactions by Product Category

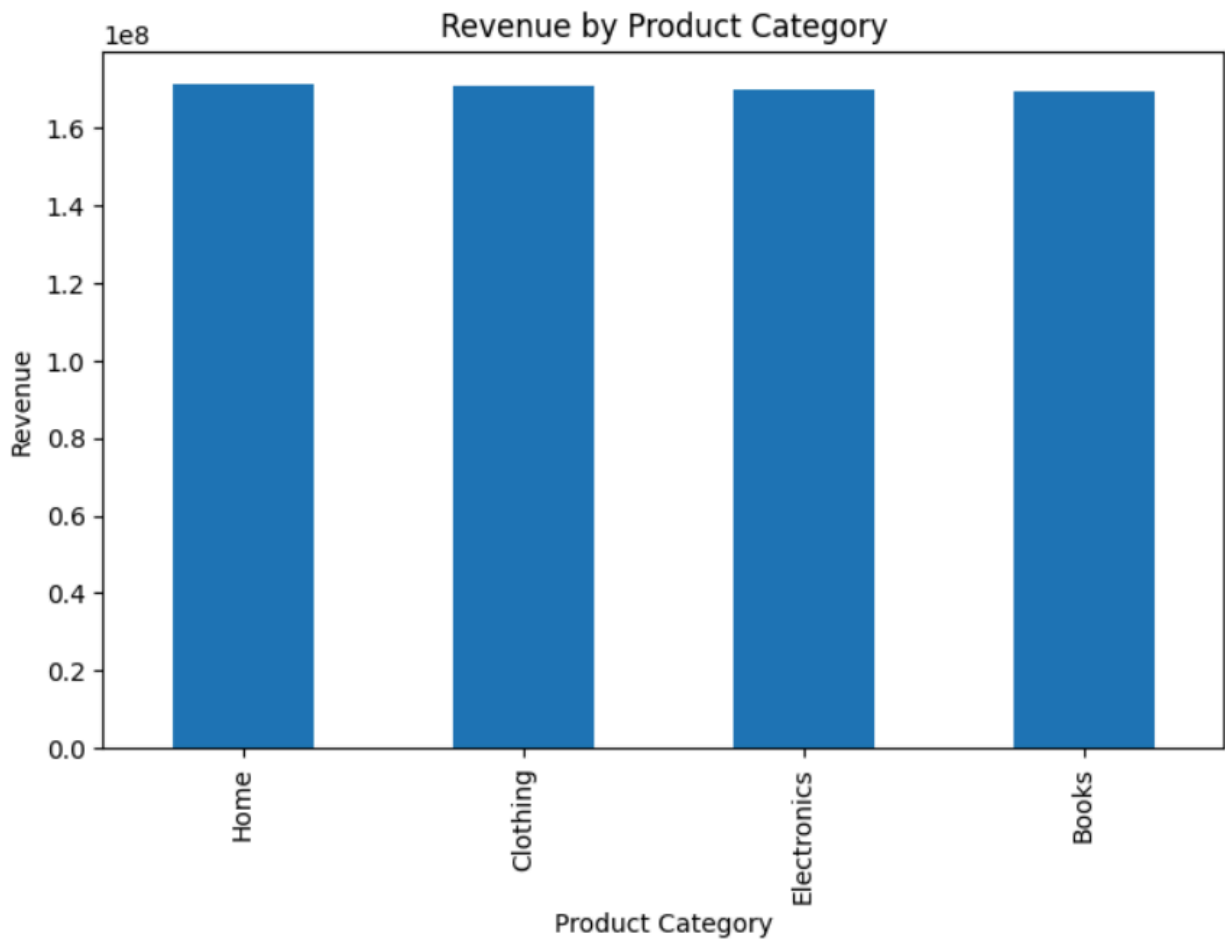


Figure 4: Revenue by Product Category

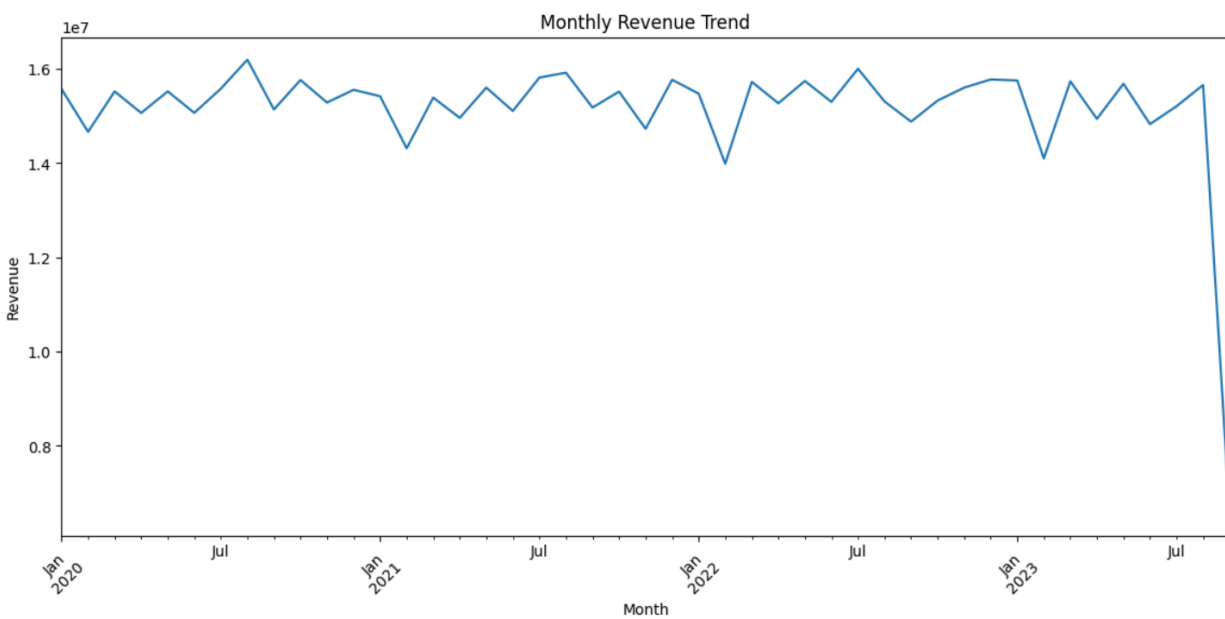


Figure 5: Monthly Revenue Trend

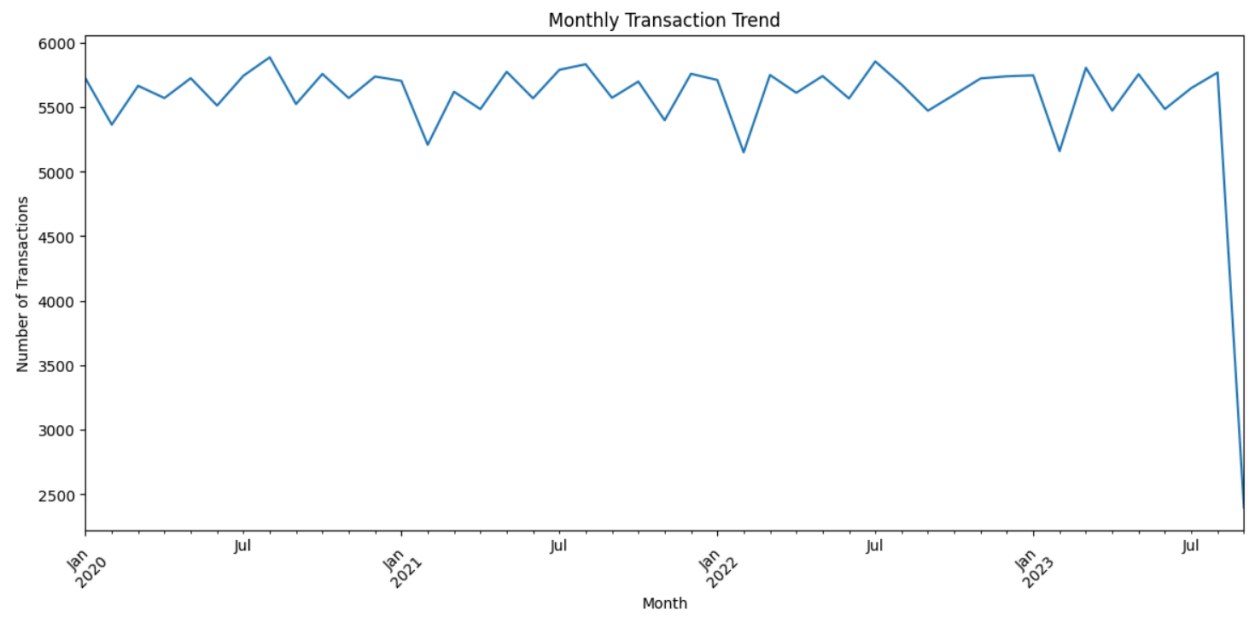


Figure 6: Monthly Transaction Trend

4. RFM Customer Segmentation

RFM (Recency, Frequency, Monetary) analysis was used to segment customers based on their purchasing behavior. Recency measures how recently a customer made a purchase, Frequency measures how often they purchased, and Monetary measures the total amount spent by the customer.

Customer Segment Profile

Segments	Customers	Avg Recency	Avg Frequency	Avg Monetary
Champions	9583	76.81	7.89	22506.30
Loyal Customers	13337	175.51	6.00	16566.33
Potential Loyalist	13357	240.15	4.31	11394.37
At Risk	7102	370.88	3.26	8306.55
Lost Customers	6282	654.15	2.18	5338.36

Potential Loyalists represent the largest customer segment with 13,357 customers. Champions have the lowest average recency of 76.81 days, the highest average frequency of 7.89 purchases, and the highest average monetary value of 22,506.30, making them the most valuable segment.

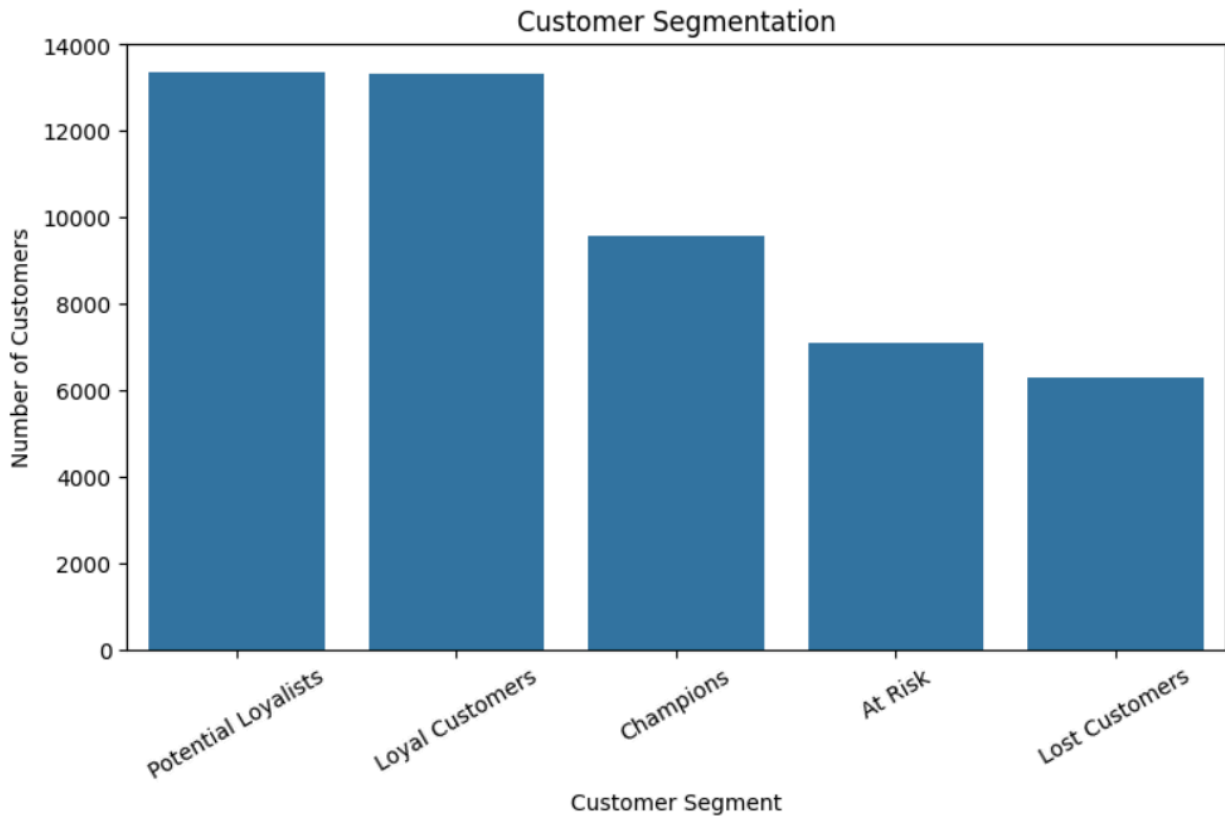


Figure 7: Customer Segmentation

The segmentation analysis shows that Potential Loyalists are the largest customer segment, followed by Loyal Customers. Champions represent a smaller but highly valuable segment, while At Risk and Lost Customers represent important groups for retention and re-engagement efforts.

5. Churn & Retention Analysis

Churn analysis was performed to understand the proportion of customers who are at risk of leaving and to identify which customer segments require greater retention efforts.

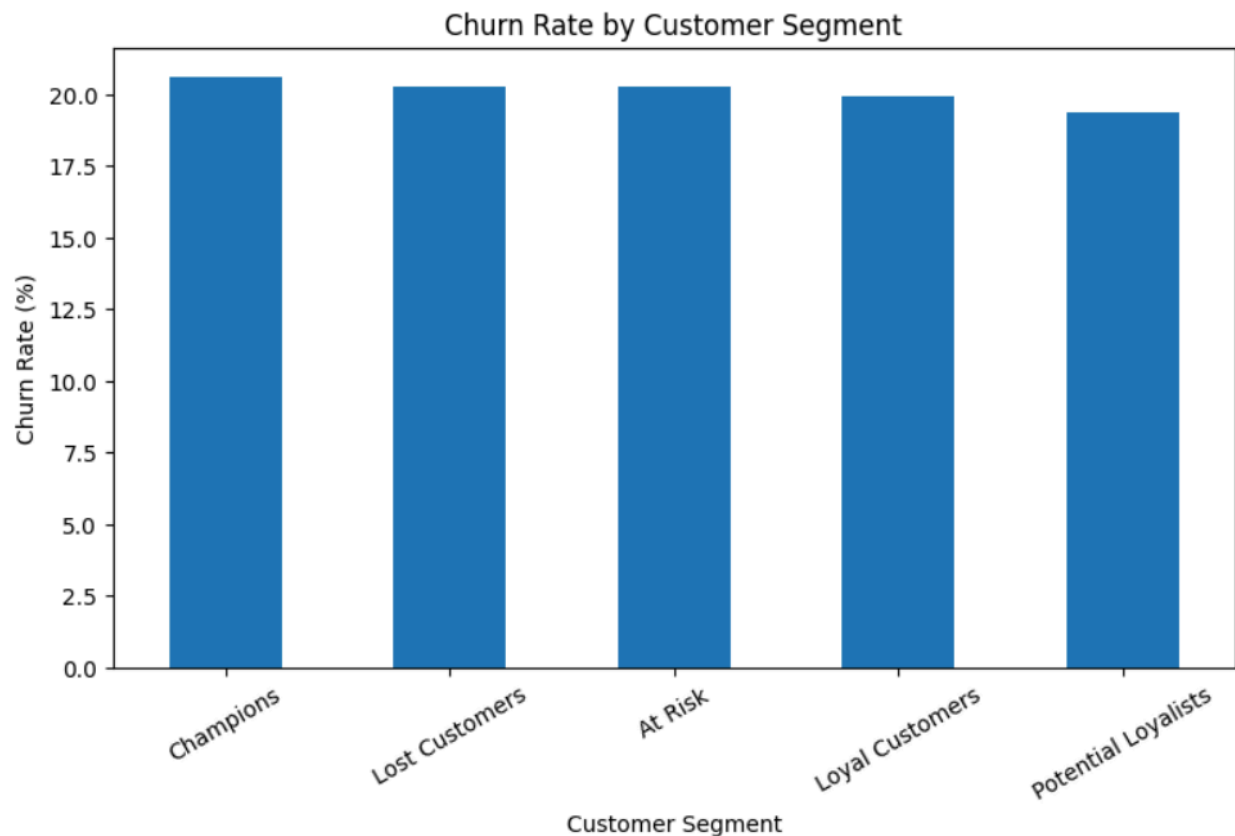


Figure 8: Churn Rate by Customer Segment

Champions have the highest churn rate at 20.59%, followed by Lost Customers at 20.28% and At Risk customers at 20.25%. Although Champions are the most valuable customer segment, their relatively high churn rate indicates a significant retention opportunity for Alfidio Tech.

Repeat Customer Rate

The analysis shows a repeat customer rate of 96.67%, indicating that the majority of customers have made more than one purchase. This suggests strong repeat-purchase behavior and provides an opportunity for Alfidio Tech to further strengthen customer loyalty.

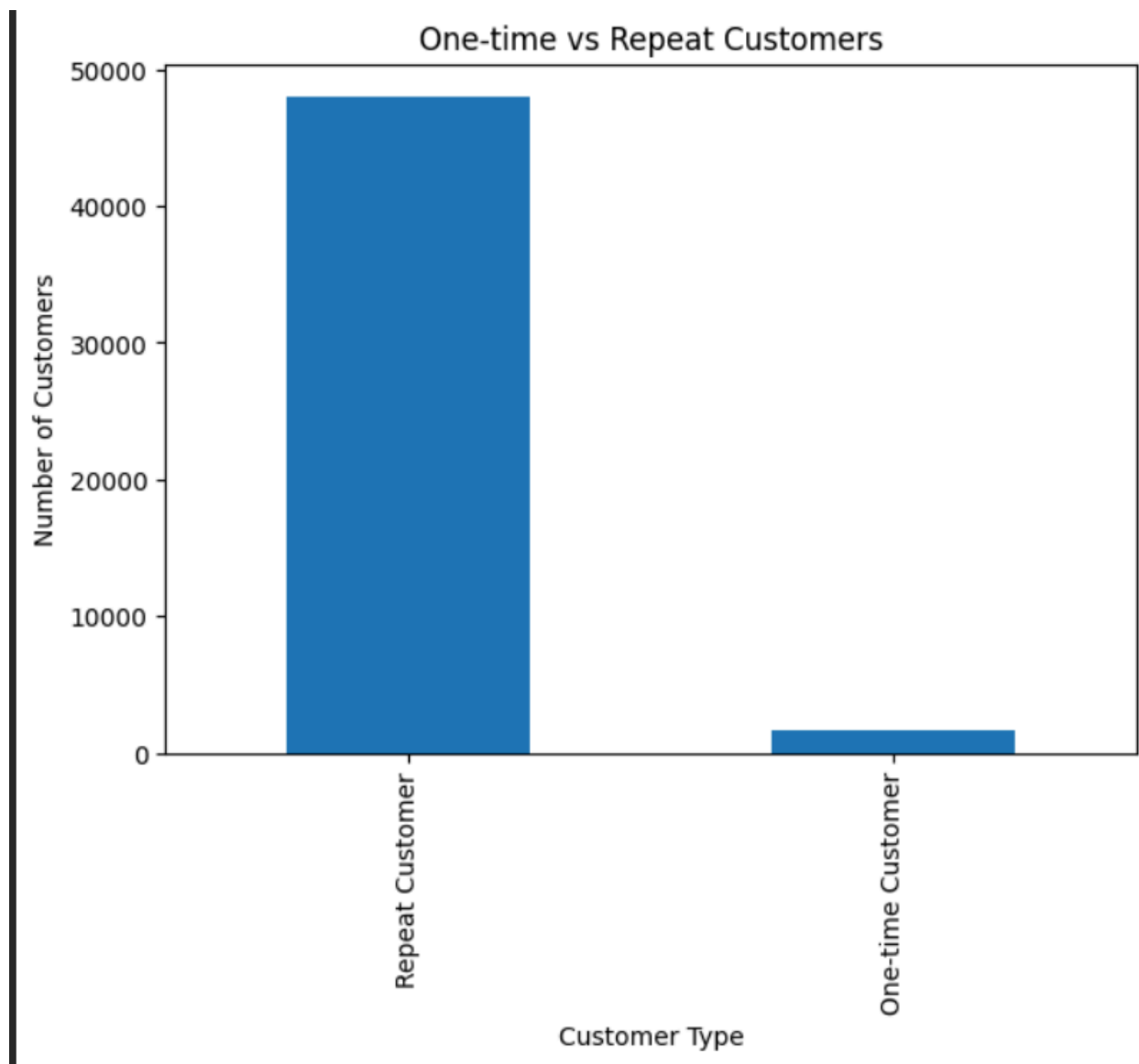


Figure 9: One-time vs Repeat Customers

One-Time vs Repeat Customers Analysis

The analysis of one-time and repeat customers helps identify customer retention patterns. Repeat customers indicate stronger customer loyalty and contribute more consistently to business revenue, while one-time customers represent potential opportunities for retention strategies. Increasing repeat purchases through personalized offers and engagement campaigns can improve customer lifetime value.

7. Key Business Insights

- Potential Loyalists are the largest customer segment with **13,357 customers**.
- Champions have the highest average monetary value at **22,506.30**.
- Champions also have the highest churn rate at **20.59%**.
- Loyal Customers have an average frequency of **6 purchases**.
- Repeat customer rate is **96.67%**.
- There is an opportunity to convert Potential Loyalists into higher-value segments

8. Recommendations for Alfido Tech

1. Protect Champion Customers

Use personalized loyalty rewards and exclusive benefits to reduce churn among high-value Champions.

2. Convert Potential Loyalists

Use targeted promotions, loyalty points and personalized recommendations to move Potential Loyalists toward Loyal Customers.

3. Strengthen Loyal Customer Engagement

Use cross-selling, personalized offers and loyalty programs to increase their purchase frequency.

4. Run Win-Back Campaigns

Target At Risk and Lost Customers with personalized re-engagement campaigns and limited-time offers.

5. Use Segment-Based Personalization

Apply different marketing strategies to different RFM segments rather than treating all customers equally.

9. Conclusion

The analysis demonstrates that RFM-based customer segmentation can help Alfido Tech understand customer value and engagement. The customer base contains a large number of Potential Loyalists and Loyal Customers, while Champions represent the highest-value group. However, the high churn rate among Champions highlights the importance of protecting valuable customers. By implementing segment-specific engagement, loyalty, and win-back strategies, Alfido Tech can improve customer retention and maximize long-term customer value.